

Sparky Build Sequence

Complete, connection-by-connection breadboard wiring instructions

This is the full build order for wiring Sparky's electronics on the breadboard, one connection at a time. It assumes the ESP32-WROOM is kept OFF the breadboard and sitting beside it - most ESP32 dev boards are too wide for a half-size 400-point board, so every ESP32 connection uses its own jumper wire rather than the board plugging in directly.

Jumper wire types matter: use a MALE-TO-FEMALE jumper for anything connecting to the ESP32 (female end on the ESP32's pin, male end into the breadboard). Use a MALE-TO-MALE jumper for anything that connects two points that are both already on the breadboard (rail-to-rail, or a component's row to a rail).

Breadboard row numbers below are a concrete suggested layout so these instructions are unambiguous - not a hardware requirement. What actually matters is that each pin of a module lands in its own row, so it can be jumpered individually. Adjust the numbers to fit your actual board if the layout doesn't match.

PHASE 0: Before You Start - Physical Prep

- 0.1. Solder the loose 6-pin header onto the INMP441 microphone module, if it didn't ship pre-soldered - check the module on arrival.
- 0.2. Solder two ~15cm wire leads onto the push button's two terminals (these are 2.8mm spade-style terminals, not breadboard pins), or crimp on 2.8mm female quick-disconnect spade connectors instead if you'd rather keep it solder-free and removable.
- 0.3. Check the MAX7219 dot matrix module on arrival: if it has its own pin header on the board edge, it can go straight into the breadboard like the other modules. If it only has the small included 5-wire cable, you'll be plugging jumper wires onto that cable's connector instead - either way, the pin names and connections below are the same.
- 0.4. Confirm you have: the half-size breadboard, both jumper wire packs (male-female and male-male), one resistor in the 150-330 ohm range from your own stash, a small soldering iron, and the ESP32 itself.

PHASE 1: Breadboard Power Rails

- 1.1. Orient the breadboard so both rail pairs (top and bottom edges) are accessible. The top rail pair will be 5V/GND, the bottom rail pair will be 3.3V/GND.
- 1.2. MALE-TO-MALE jumper: bridge the top rail's GND line to the bottom rail's GND line, so there is one shared ground plane across the whole board.
- 1.3. MALE-TO-FEMALE jumper: ESP32 '5V' or 'VIN' pin -> top rail, + line.
- 1.4. MALE-TO-FEMALE jumper: ESP32 'GND' pin -> top rail, - line.
- 1.5. MALE-TO-FEMALE jumper: ESP32 '3V3' pin -> bottom rail, + line.

Double-check with a multimeter (or just visually trace the wires) that the bottom rail is genuinely 3.3V, not 5V, before continuing - the mic in Phase 3 will be damaged if this is wrong.

PHASE 2: Place Components on the Breadboard

- 2.1. MAX98357A amp: insert its pin header into rows 1-6, one pin per row (VIN, GND, SD, BCLK, LRC,

DIN in whichever order the module's own silkscreen lists them). If your specific board also breaks out a GAIN pin, give it row 7 too - not all cheap MAX98357A boards expose this pin, so check your module first.

- 2.2. MAX7219 dot matrix: insert its header (or its included cable's connector) into rows 8-12, one pin per row (VCC, GND, CLK, DIN, CS).
- 2.3. INMP441 mic: insert its newly-soldered header into rows 14-19, one pin per row (VDD, GND, L/R, SCK, WS, SD).
- 2.4. Plain white LED: insert its two legs into rows 21 and 22 - note which leg is which: the LONGER leg is the anode (+), the SHORTER leg is the cathode (-). If the legs have been trimmed level, look for a flat spot on the LED's plastic dome, which marks the cathode side.
- 2.5. Insert your chosen resistor (150-330 ohm) so one leg shares row 21 (the LED's anode row) and the other leg sits in a fresh row, e.g. row 23.

PHASE 3: Power Connections (Male-to-Male jumpers, breadboard-to-breadboard)

- 3.1. Amp VIN row -> top rail, + line (5V).
- 3.2. Amp GND row -> top rail, - line (GND).
- 3.3. If your amp module breaks out a GAIN pin: GAIN row -> top rail, - line (GND) - sets 12dB gain (louder) instead of the ~9dB default when left unconnected. Idea from <https://www.phippselectronics.com/i2s-audio-on-the-esp32-s3/>
- 3.4. Matrix VCC row -> top rail, + line (5V).
- 3.5. Matrix GND row -> top rail, - line (GND).
- 3.6. Mic GND row -> top rail, - line (GND).
- 3.7. Mic L/R row -> top rail, - line (GND) - selects the left/mono channel.
- 3.8. Mic VDD row -> BOTTOM rail, + line (3.3V) ONLY.
Never connect the mic's VDD to the 5V rail - the INMP441's datasheet is explicit that 5V permanently destroys the chip.
- 3.9. LED's resistor row (row 23) -> top rail, + line (5V).

PHASE 4: Signal Connections (Male-to-Female jumpers, ESP32 to breadboard)

- 4.1. ESP32 GPIO 32 -> mic's SCK row (bit clock).
- 4.2. ESP32 GPIO 25 -> mic's WS row (word select).
- 4.3. ESP32 GPIO 33 -> mic's SD row (data out).
- 4.4. ESP32 GPIO 27 -> amp's BCLK row.
- 4.5. ESP32 GPIO 26 -> amp's LRC row.
- 4.6. ESP32 GPIO 22 -> amp's DIN row.
- 4.7. ESP32 GPIO 18 -> matrix's CLK row.
- 4.8. ESP32 GPIO 23 -> matrix's DIN row.
- 4.9. ESP32 GPIO 5 -> matrix's CS/LOAD row.
- 4.10. ESP32 GPIO 4 -> LED's cathode row (row 22). Remember: this pin drives the LED as a low-side switch - GPIO LOW turns the LED ON, GPIO HIGH turns it OFF (inverted from the usual setup).
- 4.11. ESP32 GPIO 13 -> a short loose wire or small exposed pad, for the capacitive touch pad. No other component needed - just an exposed conductor for a finger to touch.
- 4.12. ESP32 GPIO 21 -> amp's SD row. This replaces the old always-on wiring: SD is now under firmware control instead of hardwired to 3.3V. Firmware should drive this pin HIGH to enable the amp and LOW to mute it - in particular, mute the amp while the mic is actively recording, so the speaker's own output doesn't bleed into the mic (mic and speaker share one small enclosure on Sparky, so this matters more here than it would on a more spread-out board).

PHASE 5: Push Button

- 5.1. Button leg 1's soldered wire -> MALE-TO-FEMALE jumper -> ESP32 GPIO 14.
- 5.2. Button leg 2's soldered wire -> MALE-TO-FEMALE jumper -> ESP32 GND (or into any hole on the breadboard's GND rail, whichever is easier to reach).
No resistor needed here - the firmware will enable the ESP32's internal pull-up on GPIO 14, so the pin reads HIGH normally and LOW when the button is pressed.

PHASE 6: Speaker

- 6.1. Connect the speaker's two wires directly to the MAX98357A module's speaker output terminal (its own screw terminal or JST connector) - this does not go through the breadboard at all.

PHASE 7: Pre-Power-On Checklist

1. INMP441 VDD is on the 3.3V rail - NOT 5V. Check this one twice.
2. LED polarity is correct: anode (long leg) toward the resistor/5V side, cathode (short leg) toward GPIO 4.
3. The two GND rail pairs (top and bottom) are bridged together with a male-to-male jumper.
4. No bare wire ends are touching or bridging adjacent rows - a quick visual scan down each side of the breadboard.
5. The amp's SD pin is connected to GPIO 21, not left floating or wired to a rail - firmware must drive it HIGH to hear anything.
6. All ESP32 connections use male-to-female jumpers (female end on the ESP32).

Once everything above checks out, power the ESP32 via USB-C from the power bank. Test one peripheral at a time in firmware rather than all at once - confirm the mic transcribes, then the amp/speaker plays audio, then the matrix lights up, then the LED and button - so any wiring mistake is easy to isolate to a single connection.

GPIO pin quick reference:

GPIO 32: Mic SCK

GPIO 33: Mic SD

GPIO 26: Amp LRC

GPIO 18: Matrix CLK

GPIO 5: Matrix CS

GPIO 13: Touch pad

GPIO 21: Amp SD (mute control - HIGH = audio on)

GPIO 25: Mic WS

GPIO 27: Amp BCLK

GPIO 22: Amp DIN

GPIO 23: Matrix DIN

GPIO 4: LED cathode (low-side switch)

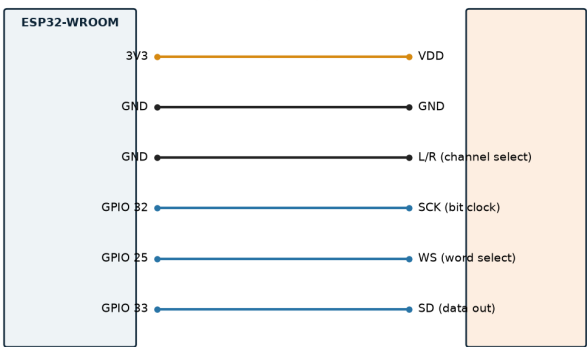
GPIO 14: Push button (internal pull-up)

Reference: Wiring Diagram

Sparky Wiring Diagram - per-component pin connections (v2, includes push-to-talk button)

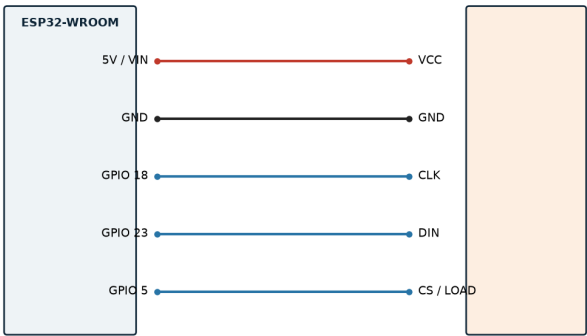
INMP441 Microphone

3.3V rail ONLY - connecting VDD to 5V will destroy the chip



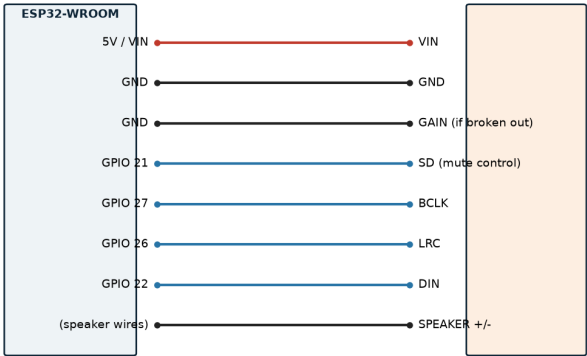
MAX7219 Dot Matrix (32x8)

5V rail - if display is flaky, power from 3.3V instead or add a level shifter



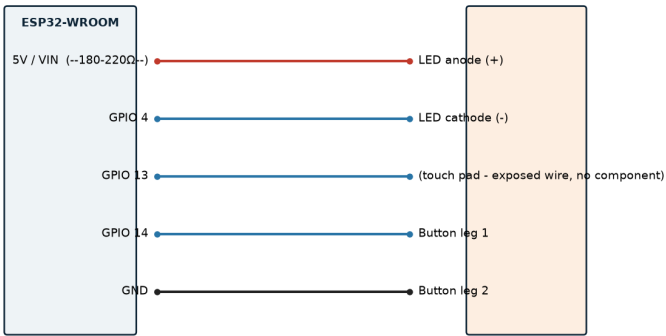
MAX98357A Amp + Speaker

5V rail. SD now GPIO-controlled (mute) - see build sequence notes



Accent LED + Touch Pad + Push-to-Talk Button

LED: plain 5mm white, low-side switched - GPIO LOW = on, HIGH = off. Button: internal pull-up



5V 3.3V GND GPIO signal